PROBLEM STATEMENT · DATA TO ACTION

Supply Chain Risk Detection & Decision-Support System

A system that turns operational data into prioritised risks, accountable owners and timely action.

CLIENT	REGION	STACK
Savanna Distribution Ltd	East Africa	Databricks · Power BI · n8n

The problem is not a lack of data. It is turning data into timely decisions.

Savanna Distribution Ltd manages products across multiple warehouses, suppliers and customers. As the business grows, so does its data, yet teams still learn about problems only after they have hit the business.

Silent stockouts

Products run out before procurement sees inventory turning critical.

Drifting suppliers

Delivery performance erodes, noticed only after several late deliveries.

Late backorders

Customer orders go backordered and the team finds out too late.

Imbalanced stock

One warehouse overstocks while another faces shortages.

No clear priority

Many risks coexist, with no way to decide which comes first.

No clear owner

Even when a risk is found, who should act is unclear.

> Which supply-chain risks need attention right now, how serious are they, what could they cost, and who needs to act?

02 KEY BUSINESS QUESTIONS

What the system must answer.

Inventory

Which products may stock out? How many days of cover remain, and will it last until replenishment? Which stockouts cost most? Where is stock excessive?

Suppliers

Who is deteriorating against their baseline? Who is chronically late? Which open purchase orders are exposed to delay?

Customer Orders

Which orders are late or backordered, carry significant financial impact, or need immediate intervention?

Decision-making

What is handled first? What is the financial exposure? Who owns each risk, and what action should they take?

03 EVIDENCE FROM THE RAW DATA

It starts with the data, not a dashboard.

168,224
deduplicated order lines

KES 967M
total order value

6
intentional data-quality issues

- Duplicate records
- Inconsistent order statuses
- Incorrectly formatted dates
- Delivered orders without delivery dates
- Invalid purchase order dates
- Negative inventory values

04 SUCCESS CRITERIA

The system must be able to:

1. Identify meaningful risks from historical and current data.
2. Estimate the financial impact of each risk.
3. Assign an urgency score.
4. Prioritise risks into P1, P2 and P3.
5. Assign each risk to the right business owner.
6. Recommend an action.
7. Present risks in a management-facing Power BI dashboard.
8. Notify owners through the automation layer.

> The goal is not the maximum number of alerts. It is useful alerts that help people decide what to do next.

05 BUSINESS OUTCOME

Reactive problem solving
Discover the stockout after it happens.

→

Proactive risk management
Spot the risk early enough to respond.

06 FROM DATA TO ACTION

One decision flow, eight layers.

01
Data
What happened.

02
Evidence
Raw data into business signals.

03
Risk
Potential problems.

04
Impact
What it could cost.

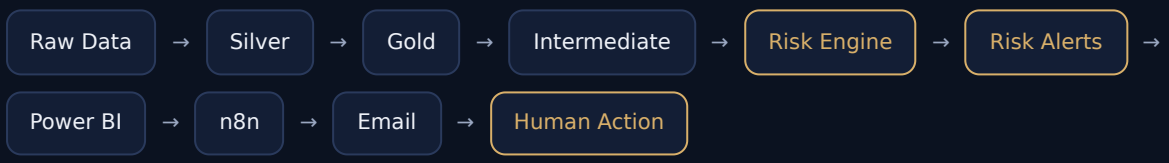
05
Urgency
How fast to respond.

06
Priority
What to handle first.

07
Owner
Who is responsible.

08
Action
A practical next step.

07 SOLUTION ARCHITECTURE



Databricks
Processes data, builds business-ready datasets, calculates risk indicators and generates prioritised alerts.

Power BI
The Supply Chain Risk Command Centre for monitoring the live risk landscape.

n8n
The automation layer that routes risk information to the right owner.

Email
The operational notification, so risks never stay hidden in a dashboard.

> Don't build a dashboard just to show data. Build a system that helps someone make a decision.